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Energy and Buildings

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About the journal

Aims and scope

An international journal devoted to investigations of energy use and efficiency in buildings

Energy and Buildings is an international journal publishing articles with explicit links to energy use in buildings. The aim is to present new research results, and new proven practice aimed at reducing the energy needs of a building and improving indoor environment quality. Energy and Buildings considers and publishes articles considerably advancing Building Science. Preference is given to practical and experimental research articles reporting considerable innovations. Articles reporting advances in theoretical and simulation methods are welcome if results are validated, calibrated, or benchmarked using appropriate experimental or well-recognized data, and so are the articles reporting applications of numerical or theoretical methods for the analysis of new technologies, materials, and innovative designs or operations.

Topics covered include:

- Energy demands and consumption in existing, new, and future buildings – prediction, benchmarking, and validation
- Indoor environment quality, including safety, health, thermal and visual comfort vis-à-vis energy
- Natural, mechanical and hybrid ventilation
- Air distribution in buildings
- Application of solar and other renewable energy sources in buildings
- Energy balances in building complexes (residential, commercial, industrial, public and other buildings)
- Energy efficiency improvement measures of HVAC&R and other technical systems in residential, commercial, public and industrial buildings, and semi open built spaces
- Heat recovery systems in buildings
- Buildings and district heating and cooling
- Energy conservation in built environment
- Energy efficient buildings
- Building physics
- Energy sustainability, resilience and climate adaptability of buildings
- Evaluation and control of indoor thermal and lighting systems

- Building's total performance and influencing factors
- Intelligent/smart buildings and communities
- Links between architectural design, mechanical and lighting systems
- New materials in buildings and their impact on energy demands
- External and internal design conditions for energy efficient buildings
- Building envelope materials and structure energy performance
- Thermal energy storage and thermally active building systems - TABS
- Energy performance of buildings
- Advanced controls of energy systems in buildings: model predictive controls, AI-based controls
- Zero CO2 emission - zero energy and energy plus buildings and their smart grid harmonized operation
- Residential/municipal energy refurbishment and renovation
- Life cycle energy efficiency of buildings and embodied energy
- Architectural structure - construction energy efficiency
- Energy related aspects of buildings after catastrophic events
- Applications of AI related to energy use in buildings

- Technological issues in energy poverty in building sectors
- Impacts of urban microclimate and wind-structure interaction on the thermal comfort in building surroundings, building ventilation and energy use
- Global and regional climate change and its potential impact on the energy and environmental quality of the built environment
- New heat and energy mitigation and adaptation technologies for the built environment
- Successful case studies demonstrating the implementation of innovative energy and environmental technologies in the built environment
- Synergies between energy consumption and indoor environmental quality
- The impact of human behaviors on the energy consumption of buildings
- Distributed energy resources (behind-the-meter) in buildings
- Energy flexibility and demand response of buildings
- Grid-responsive buildings and integrated energy systems
- Interconnections between buildings, urban climate, transportation, and power grid
- Datasets of operational performance of buildings and energy systems
- Future and extreme weather data supporting energy and buildings research
- Energy use in data centers and industry facilities

- Advanced building performance modelling and simulation techniques
- Decarbonization (including electrification) of building energy systems
- Digital twins of buildings and energy systems
- Advanced analytics of buildings and systems performance
- Building energy policy, codes and standards

Papers with results based on simulations are welcome but those with clear links to laboratory or field measurements are preferred. These links may include calibration, benchmarking, validation, or comparisons of results.

Replication Studies: This journal encourages the submission of replication studies. When submitting your manuscript ensure the original study is clearly referenced, and provide a link to the original study. Replication studies should include [brief] introduction and discussion sections that succinctly report the goal of the original paper. The original paper should be the work of a different author or group of authors. For replication studies please also deposit your data in Mendeley data or another trusted repository and ensure your data is properly cited. Please refer to the section on Research data for further guidance on data deposition and citation.

Please follow and tag the journal on one of our Social media accounts if you are sharing a link to your paper on ScienceDirect: [Twitter](#) (@ENBJournal) and [LinkedIn](#).

Article types

Original papers

(preferably no more than 20 double line spaced manuscript pages including tables and illustrations),

Review Articles and Short Communications

(no more than 10 double line spaced manuscript pages including tables and figures).

Authors should produce a manuscript that can be reviewed on its own. Authors considering the submission of multiple papers (Part 1, Part 2, for example) are advised to combine them into one paper. If authors choose to submit the respective parts sequentially, they should not assume that later parts will be published. Each submission will be considered on its own merits. Text accompanying figures should be reasonably brief and not extensive.

Discussions

Any comment on or discussion of a published paper should be emailed directly to the Editor, NOT submitted via the electronic submission system. It must be received by the Editor within three months of the online publication of the paper under discussion.

Replication Studies:

This journal encourages the submission of replication studies. When submitting your manuscript ensure the original study is clearly referenced, and provide a link to the original study. Replication studies should include [brief] introduction and discussion sections that succinctly report the goal of the original paper. The original paper should be the work of a different author or group of authors. For replication studies please also deposit your data in Mendeley data or another trusted repository and ensure your data is properly cited. Please refer to the section on Research data for further guidance on data deposition and citation.

Peer review

This journal follows a single anonymized review process. Your submission will initially be assessed by our editors. If your submission is deemed suitable, it will typically be sent to be reviewed by a minimum of two reviewers for an independent expert assessment of the scientific quality. The decision as to whether your article is accepted or rejected for publication will be made by our editors.

Read more about [peer review](#).

Our editors are not involved in making decisions about papers which:

- they have written themselves.
- have been written by family members or colleagues.
- relate to products or services in which they have an interest.

Any such submissions will be subject to the journal's usual procedures and peer review will be handled independently of the editor involved and their research group. Read more about [editor duties](#).

Authors may submit a formal appeal request to the editorial decision, provided it meets all the requirements and follows the procedure outlined in Elsevier's [Appeal Policy](#). Only one appeal per submission will be considered and the appeal decision will be final.

Special issues and article collections

The peer review process for special issues and article collections follows the same process as outlined above for regular submissions, except, a guest editor may send the submissions out to the reviewers and may recommend a decision to the journal editor. The journal editor oversees the peer review process of all special issues and article collections to ensure the high standards of publishing ethics and responsiveness are respected and is responsible for the final decision regarding acceptance or rejection of articles.

Open access

We refer you to our [open access information page](#) to learn about open access options for this journal.

Ethics and policies

Ethics in publishing

Authors must follow ethical guidelines stated in Elsevier's [Publishing Ethics Policy](#).

Submission declaration

When authors submit an article to an Elsevier journal it is implied that:

- The work described has not been published previously except in the form of a preprint, an abstract, a published lecture, academic thesis or registered report. See our policy on [multiple, redundant or concurrent publication](#).
- The article is not under consideration for publication elsewhere.
- The article's publication is approved by all authors and tacitly or explicitly by the responsible authorities where the work was carried out.

- If accepted, the article will not be published elsewhere in the same form, in English or in any other language, including electronically, without the written consent of the copyright-holder.

To verify compliance with our journal publishing policies, we may check your manuscript with our screening tools.

Authorship

All authors should have made substantial contributions to all of the following:

1. The conception and design of the study, or acquisition of data, or analysis and interpretation of data
2. Drafting the article or revising it critically for important intellectual content.
3. Final approval of the version to be submitted.

All authors are accountable for all aspects of the work, to ensure that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Corresponding author

Authors must appoint a single corresponding author to communicate with the journal during the editorial process. This individual submits the manuscript and all required documentation to the journal, and serves as the primary contact between the journal and the co-authors. It is important that all contact details of the corresponding author are kept up to date throughout the submission and publication process.

Only the corresponding author's affiliation will be used to determine eligibility for a publishing agreement, and possible discounts related to it. Affiliations of other co-authors are not relevant for eligibility.

Group authorship

Group authorship refers to the attribution of an article to a defined and named group of authors whose members have contributed to the work. Examples include research consortia, clinical trial groups, collaborative networks, task forces or working groups, expert panels or consensus groups, and participatory research groups.

- Group authorship must be declared on the title page *at submission*. When available, the submitting author should include the group under “Collaborative Author Groups” in the submission system.
- A corresponding author who agrees to coordinate all communications on behalf of the group must be designated.
- The group name will be included in the author list in the published article (e.g., “on behalf of the [Consortium Name]”).
- All group members associated with the work should be listed in a dedicated section, preferably an appendix.
- All listed authors must follow the journal’s authorship and authorship change policies.
- The group must maintain a transparent internal process for determining authorship and resolving disputes. All authorship disputes must be resolved within the group.

Changes to authorship

The editors of this journal generally will not consider changes to authorship once a manuscript has been submitted. It is important that authors carefully consider the authorship list and order of authors and provide a definitive author list at submission.

The policy of this journal around authorship changes:

- All authors must be listed in the manuscript and their details entered into the submission system. Changes can only be made prior to acceptance, and only if approved by the journal editor. This includes additions, deletion, or rearrangement of author names.
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- This journal does not allow authorship changes after acceptance. This includes additions, deletions, or the rearrangement of author names, including changes to the corresponding author.

- The review process may be paused while a change in authorship request is being considered.
- Change requests made by the authors without the [Authorship Change Request form](#) or editor approval may result in the rejection of the manuscript, or retraction if the article has already been published.

Declaration of competing interests

All authors must disclose any financial and personal relationships with other people or organizations that could inappropriately influence or bias their work. Examples of potential competing interests include:

- Employment
- Consultancies
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- Honoraria
- Paid expert testimony
- Patent applications or registrations
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The [declarations tool](#) should always be completed.

Authors with a journal affiliation to declare should enter the following text under “Other Activities” within the [declarations tool](#) and should inform the journal and publisher prior to completing the submission process:

Given their role as [AUTHOR JOURNAL ROLE TITLE], [AUTHOR NAME] had no involvement in the peer-review of this article and has no access to information regarding its peer-review. Full responsibility for the editorial process for this article was delegated to another journal editor.

Editorial disclosure statements will be included as a footnote and/or in the declaration of competing interest section of the article.

Authors with no competing interests to declare should select the option "I have nothing to declare".

The resulting Word document containing your declaration should be uploaded at the "attach/upload files" step in the submission process. It is important that the Word document is saved in the .doc/.docx file format. Author signatures are not required.

Funding sources

Authors must disclose any funding sources who provided financial support for the conduct of the research and/or preparation of the article. The role of sponsors, if any, should be declared in relation to the study design, collection, analysis and interpretation of data, writing of the report and decision to submit the article for publication. If funding sources had no such involvement this should be stated in your submission.

List funding sources in this standard way to facilitate compliance to funder's requirements:

Funding: This work was supported by the National Institutes of Health [grant numbers xxxx, yyyy]; the Bill & Melinda Gates Foundation, Seattle, WA [grant number zzzz]; and the United States Institutes of Peace [grant number aaaa].

It is not necessary to include detailed descriptions on the program or type of grants, scholarships and awards. When funding is from a block grant or other resources available to a university, college, or other research institution, submit the name of the institute or organization that provided the funding.

If no funding has been provided for the research, it is recommended to include the following sentence:

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Authors must declare the use of generative AI tools in the manuscript preparation process upon submission of the paper.

Generative AI and AI-assisted technologies (“AI tools”) have been rapidly adopted in research and scholarly publishing, offering significant benefits while requiring clear standards for responsible and transparent use.

Authors preparing a manuscript for this journal may use AI tools to support them. However, these tools must never be used as a substitute for human critical thinking, expertise and evaluation. AI tools may only be applied with human oversight and control.

Please read Elsevier’s detailed author policy on the use of AI tools, as stated in Elsevier’s [generative AI policies for journals](#).

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- Carefully reviewing and verifying the accuracy, comprehensiveness, and impartiality of all AI-generated output (including checking the sources, as AI-generated references can be incorrect or fabricated).
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- Ensuring accuracy and originality of any AI generated images and provide appropriate attribution where relevant.
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- Ensuring the manuscript is developed in a way that safeguards data privacy, intellectual property and other rights, by checking the terms and conditions of any AI tool that is used.

Finally, authors must not list or cite AI tools as an author or co-author on the manuscript since authorship implies responsibilities and tasks that can only be attributed to, and performed by, humans.

The use of AI tools in the manuscript preparation process must be declared by adding a statement at the end of the manuscript when the paper is first submitted. The statement will appear in the published work and should be placed in a new section before the references list. An example:

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Statement: *During the preparation of this work the author(s) used [NAME OF TOOL / SERVICE] in order to [REASON]. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.*

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It is expected that the corresponding author will seek approval from all co-authors before agreeing to post the manuscript publicly, prior to peer review, on SSRN.

Use of inclusive language

Inclusive language acknowledges diversity, conveys respect to all people, is sensitive to differences, and promotes equal opportunities. Authors should ensure their work uses inclusive language throughout and contains nothing which might imply one individual is superior to another on the grounds of:

- age
- gender
- race
- ethnicity

- culture
- sexual orientation
- disability or health condition

We recommend avoiding the use of descriptors about personal attributes unless they are relevant and valid. Write for gender neutrality with the use of plural nouns ("clinicians, patients/clients") as default. Wherever possible, avoid using "he, she," or "he/she."

No assumptions should be made about the beliefs of readers and writing should be free from bias, stereotypes, slang, reference to dominant culture and/or cultural assumptions.

These guidelines are meant as a point of reference to help you identify appropriate language but are by no means exhaustive or definitive.

Reporting sex- and gender-based analyses

There is no single, universally agreed-upon set of guidelines for defining sex and gender. We offer the following guidance:

- Sex and gender-based analyses (SGBA) should be integrated into research design when research involves or pertains to humans, animals or eukaryotic cells. This should be done in accordance with any requirements set by funders or sponsors and best practices within a field.
- Sex and/or gender dimensions of the research should be addressed within the article or declared as a limitation to the generalizability of the research.
- Definitions of sex and/or gender applied should be explicitly stated to enhance the precision, rigor and reproducibility of the research and to avoid ambiguity or conflation of terms and the constructs to which they refer.

We advise you to read the [Sex and Gender Equity in Research \(SAGER\) guidelines](#) and the [SAGER checklist \(PDF\)](#) on the EASE website, which offer systematic approaches to the use of sex and gender information in study design, data analysis, outcome reporting and research interpretation.

For further information we suggest reading the rationale behind and recommended [use of the SAGER guidelines](#).

Definitions of sex and/or gender

We ask authors to define how sex and gender have been used in their research and publication. Some guidance:

- Sex generally refers to a set of biological attributes that are associated with physical and physiological features such as chromosomal genotype, hormonal levels, internal and external anatomy. A binary sex categorization (male/female) is usually designated at birth ("sex assigned at birth") and is in most cases based solely on the visible external anatomy of a newborn. In reality, sex categorizations include people who are intersex/have differences of sex development (DSD).
- Gender generally refers to socially constructed roles, behaviors and identities of women, men and gender-diverse people that occur in a historical and cultural context and may vary across societies and over time. Gender influences how people view themselves and each other, how they behave and interact and how power is distributed in society.

Depending on the focus of a paper, sex and/or gender may or may not be relevant to the content of the paper. We recognize that beliefs, attitudes, and laws relating to sex and gender may vary. These articles do not attempt to dictate author beliefs but rather require that, where relevant to an author's research or paper, the author must provide clear explanations of how the paper and research define and use sex and gender.

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We accept that authors sometimes need to adjust images for clarity but any manipulation of images for the purpose of deception or fraud will be seen as scientific ethical abuse and will be dealt with accordingly.

Authors must adhere to this journal's policy for graphical images:

- No specific feature within an image may be enhanced, obscured, moved, removed or introduced.
- Adjustments of brightness, contrast, or color balance are acceptable if, and only as long as, they do not obscure or eliminate any information present in the original image.

- Nonlinear adjustments such as changes to gamma settings must be disclosed in the figure legend.
- Use of AI tools to create or manipulate primary research images is not permitted. This includes images that represent experimental or observational data, such as western blots, microscopy images, histology images, or patient images. Please read our policy on the use of AI tools in figures, images and artwork, which can be found in Elsevier's [GenAI Policies for Journals](#).

To verify compliance with the above, this journal may send your images to a secure third-party tool to enable screening for image irregularities. Our editors may ask you to provide original data or images if any questions arise as a result of the screening. The final decision as to whether images are acceptable will be taken by our editors.

Authors are encouraged to carefully check all images before submission and to connect all the data in any figures to the original, unprocessed data.

Jurisdictional claims

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- Institutional affiliations: Authors should use either the full, standard title of their institution or the standard abbreviation of the institutional name so that the institutional name can be independently verified for research integrity purposes.

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If you need to delay the publication of your article for any reason, please contact the editorial office of the journal or our [Journal Article Publishing Support Center](#) at the earliest possible opportunity. While we will try to ensure that your article is not published before a certain date if you make your request on a timely basis, we cannot guarantee this. Please also note that once the agreed period of delay has expired the article may be published at any time thereafter.

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For questions about article withdrawal, correction, or retraction, please consult Elsevier's [Article Correction, Retraction and Removal Policy](#).

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File format

We ask you to provide editable source files for your entire submission (including figures, tables and text graphics). Some guidelines:

- Save files in an editable format, using the extension .doc/.docx for Word files and .tex for LaTeX files. A PDF is not an acceptable source file.
- Format Word files in a single-column layout. Double-column formatting is only permitted for LaTeX submissions.
- Remove any strikethrough and underlined text from your manuscript, unless it has scientific significance related to your article.
- Use spell-check and grammar-check functions to avoid errors.

We advise you to read our [Step-by-step guide to publishing with Elsevier](#).

LaTeX

We encourage you use our [LaTeX template](#) when preparing a LaTeX submission. You will be asked to provide all relevant editable source files upon submission or revision.

Support for your LaTeX submission:

- [LaTeX submission instructions and templates](#)
- Journal Article Publishing Support Center [LaTeX FAQs and support](#)
- Researcher Academy's [Beginners' guide to writing a manuscript in LaTeX](#)

Title page

You are required to include the following details on the title page:

- Article title. Article titles should be concise and informative. Please avoid abbreviations and formulae, where possible, unless they are established and widely understood, e.g. DNA.
- Author names. Provide the given name(s) and family name(s) of each author. The order of authors should match the order in the submission system. Carefully check that all names are accurately spelled. You may add your name between parentheses in your own script after the English transliteration.
- Affiliations, referring to where the work was carried out, below the author names. Indicate affiliations using a lower-case superscript letter immediately after the author's name and in front of the corresponding address. Ensure that you provide the full name and postal address of each affiliation, including the country name.
- Corresponding author. Clearly indicate who will handle correspondence for your article at all stages of the refereeing and publication process and also post-publication. This responsibility includes answering any future queries about your results, data, or methodology. The email address of the corresponding author must be included on the title page and will be published in the article.
- Present/permanent address. If an author has changed affiliation since the work described in your article was carried out, or was visiting during that time, a "present address" (or "permanent address") can be indicated by a footnote to the author's name. The address where the author carried out the work must be retained as their main affiliation address. Use superscript Arabic numerals for such footnotes.

Abstract

You are required to provide a concise and factual abstract which does not exceed 250 words. The abstract should:

- State the purpose of your paper.
- Outline basic procedures followed.
- Include your main findings, their statistical significance, where applicable, and your principal conclusions.
- Emphasize new and important aspects of your study or observations.

Some guidelines:

- Abstracts must be able to stand alone, as abstracts are often presented separately from the article.
- Avoid references. If any are essential to include, ensure that you cite the author(s) and year(s).
- Avoid non-standard or uncommon abbreviations. If any are essential to include, ensure they are defined within your abstract at first mention.

Keywords

You are required to provide 1 to 7 keywords for indexing purposes. Keywords should be written in English. Please try to avoid keywords consisting of multiple words (using "and" or "of").

We recommend that you only use abbreviations in keywords if they are firmly established in the field.

Highlights

You are required to provide article highlights at submission.

Highlights are a short collection of bullet points that should capture the novel results of your research as well as any new methods used during your study. Highlights will help increase the discoverability of your article via search engines. Some guidelines:

- Submit highlights as a separate editable file in the online submission system with the word "highlights" included in the file name.
- Highlights should consist of 3 to 5 bullet points, each a maximum of 85 characters, including spaces.

We encourage you to view example [article highlights](#) and read about the benefits of their inclusion.

Graphical abstract

You are required to provide a graphical abstract at submission.

The graphical abstract summarizes the contents of your article in a concise, pictorial, professional form, highlighting important aspects of the research. It is designed to appeal to an interdisciplinary audience, and draw more attention to your online article. Some guidelines:

- Ensure you have obtained the necessary permission to include any third party owned material in a graphical abstract.
- Use of generative AI or AI-assisted tools in the production of graphical abstracts must align with Elsevier's [GenAI Policies for Journals](#).
- Submit your graphical abstract as a separate file in the online submission system.
- Ensure the image is 531 x 1328 pixels (h x w) or proportionally more, and is readable at a size of 5 x 13 cm.
- Preferred file types for graphical abstracts are TIFF, EPS, PDF or MS Office files.

More information, examples and a template can be found in the [graphical abstract guidelines](#).

Math formulae

- Submit math equations as editable text, not as images.
- Present simple formulae in line with normal text, where possible.
- Use the solidus (/) instead of a horizontal line for small fractional terms such as X/Y.

- Present variables in italics.
- Denote powers of e by exp.
- Display equations separately from your text, numbering them consecutively in the order they are referred to within your text.

Tables

Tables must be submitted as editable text, not as images. Some guidelines:

- Place tables next to the relevant text or on a separate page(s) at the end of your article.
- Cite all tables in the manuscript text.
- Number tables consecutively according to their appearance in the text.
- Please provide captions along with the tables.
- Place any table notes below the table body.
- Avoid vertical rules and shading within table cells.

We recommend that you use tables sparingly, ensuring that any data presented in tables is not duplicating results described elsewhere in the article.

Figures, images and other artwork

Figures, images, diagrams and other graphical media (artwork) must be supplied as separate files along with the manuscript, in line with these [artwork and media instructions](#). Some excerpts:

When submitting artwork:

- Cite all images in the manuscript text.
- Number images according to the order they appear within your article.
- Submit each image as a separate file using a logical naming convention for your files (for example, Figure_1, Figure_2 etc).
- Text graphics may be embedded in the text at the appropriate position. If you are working with LaTeX, text graphics may also be embedded in the file.

Captions

All artwork must have a caption. A caption should consist of a brief title (not displayed on the figure itself) and a description of the image. Keep the amount of text to a minimum, though any symbol or abbreviation should be explained.

Read how to add captions to your submission [here](#).

Formats

Submit your artwork in the formats listed below, following the given requirements:

Image type	Recommended file type	Requirements
Vector drawings	EPS, PDF	Embed the font or save text as graphics
Photographs (color or grayscale)	TIFF, JPG, PNG	Minimum resolution 300 dpi (min. width: single column: 1063 pixels; full page: 2244 pixels)

Bitmapped line drawings	TIFF, JPG, PNG	Minimum resolution 1000 dpi (min. width: single column: 3543 pixels; full page: 7480 pixels)
Charts and large or shaded tables	XLS, XLSX	Supply in a clearly readable format
Annotated images	PPT, PPTX	Annotations and text must be legible; Minimum resolution of embedded images 300 dpi

Do not submit:

- Files that are too low in resolution.
- Artwork where text is disproportionately small compared to the image size, as text may become unreadable.
- Multiple images or graphs combined into one file, as this affects legibility and accessibility. An exception may be made for images that are intended to be compared to each other, such as before, during, and after photographs.

Color artwork

If you submit usable color figures with your accepted article, we will ensure that they appear in color online. Please ensure that color images are accessible to all, including those with impaired color vision. Learn more about [color and web accessibility](#).

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Reference to a book:

[3] W. Strunk Jr., E.B. White, *The Elements of Style*, fourth ed., Longman, New York, 2000.

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[4] G.R. Mettam, L.B. Adams, How to prepare an electronic version of your article, in: B.S. Jones, R.Z. Smith (Eds.), *Introduction to the Electronic Age*, E-Publishing Inc., New York, 2020, pp. 281 - 304.

Reference to a website:

[5] Cancer Research UK, Cancer statistics reports for the UK. <http://www.cancerresearchuk.org/aboutcancer/statistics/cancerstatsreport/>, 2023 (accessed 13 March 2023).

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[6] M. Oguro, S. Imahiro, S. Saito, T. Nakashizuka, Mortality data for Japanese oak wilt disease and surrounding forest compositions [dataset], Mendeley Data, v1, 2015. <https://doi.org/10.1234/abc12nb39r.1>.

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[7] E. Coon, M. Berndt, A. Jan, D. Svyatsky, A. Atchley, E. Kikinzon, D. Harp, G. Manzini, E. Shelef, K. Lipnikov, R. Garimella, C. Xu, D. Moulton, S. Karra, S. Painter, E. Jafarov, S. Molins, *Advanced Terrestrial Simulator (ATS) v0.88* [software], Zenodo, March 25, 2020. <https://doi.org/10.1234/zenodo.3727209>.

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